Generative Score-Based Models: Reverse Engineering Noise

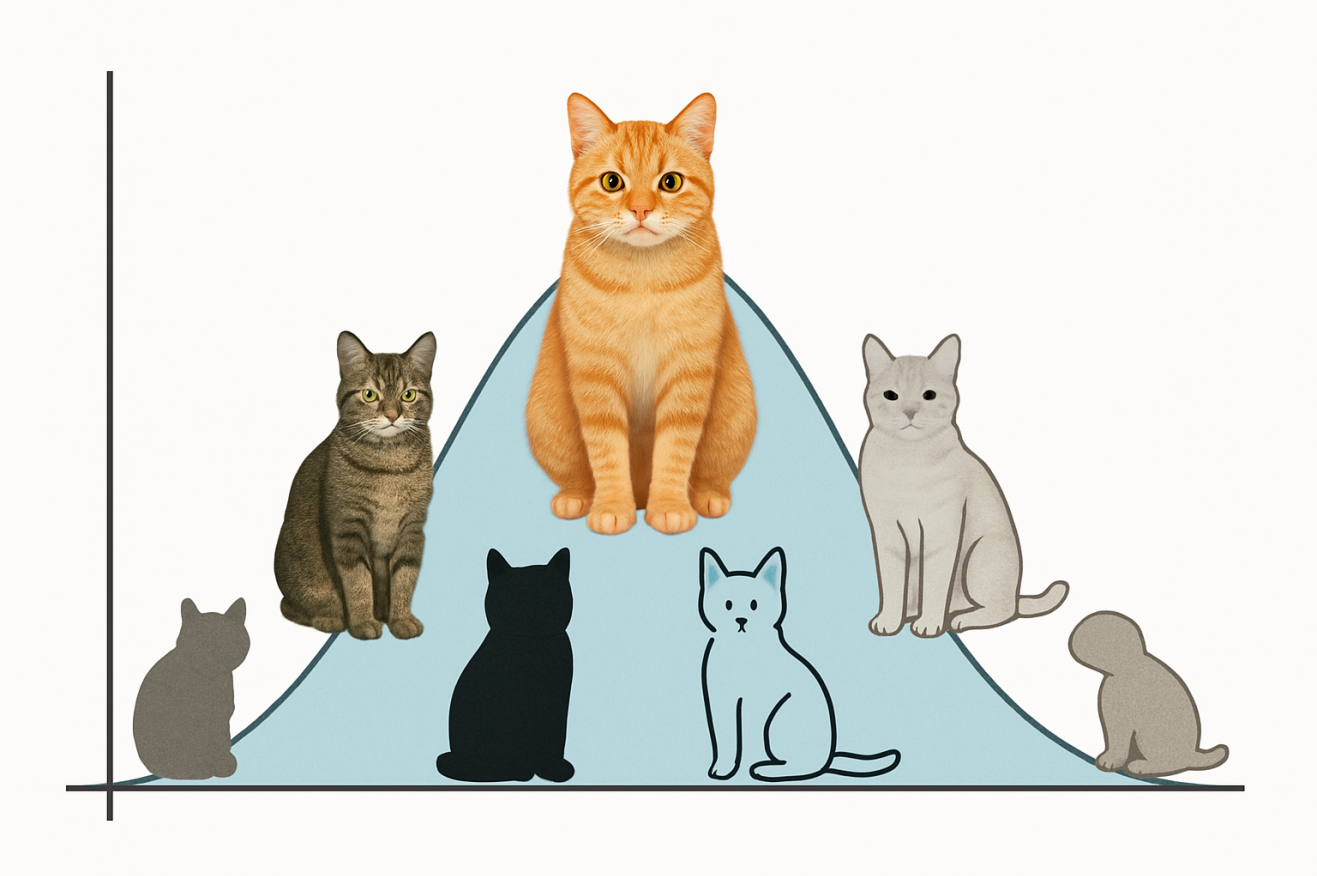
Vinay Joshy

March 31, 2025

Image Generation

Generated using ChatGPT

Probability distribution of cats



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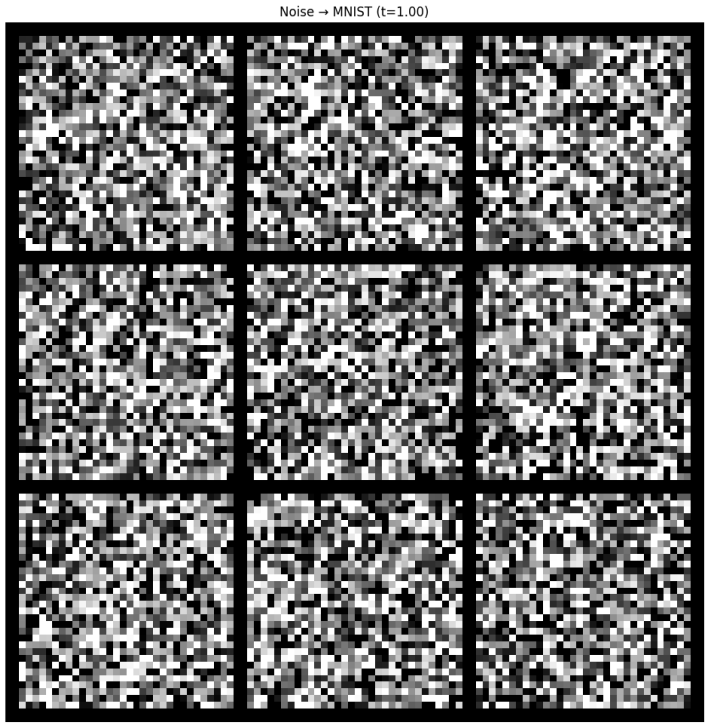

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- The partition function Z_θ makes it difficult to learn normalized models with maximum likelihood estimation.

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- **New Goal**: Estimate score function!

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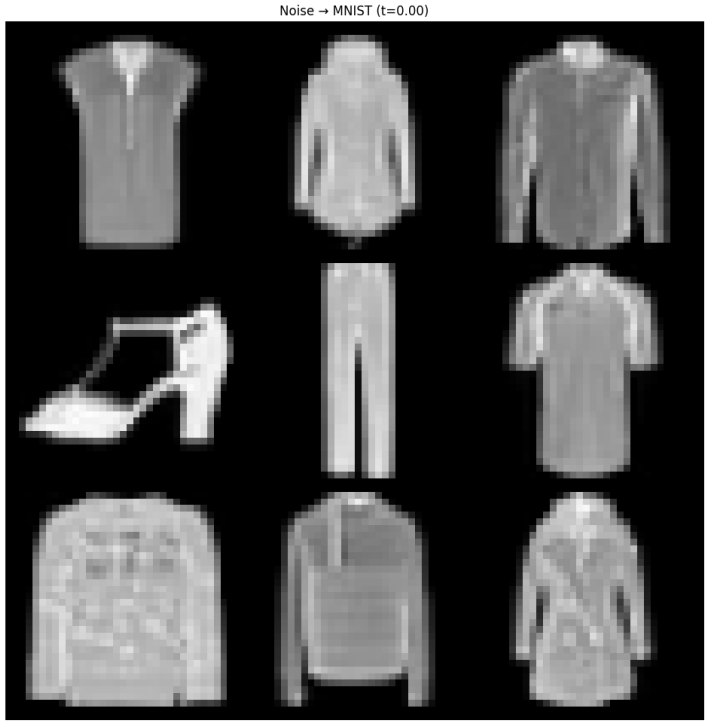

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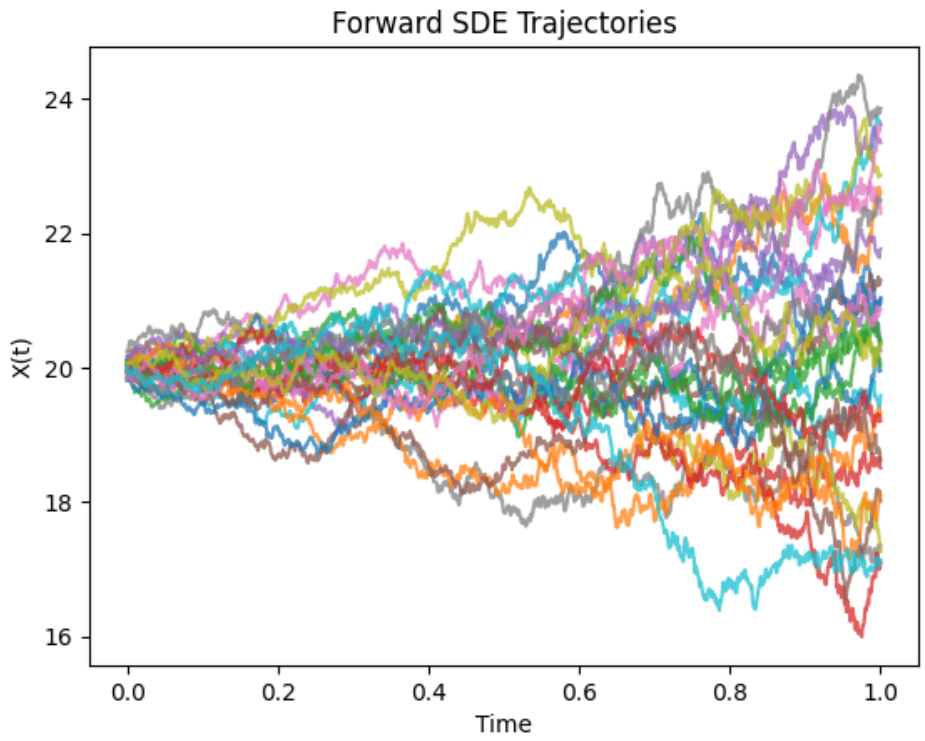
- Components:
  - $f(.,t) : \mathbb{R}^d \to \mathbb{R}$: Drift coefficient (deterministic part)
  - $g(t) \in \mathbb{R}$: Diffusion coefficient (noise amplitude)
  - dw: Infinitesimal white noise
  - w: Brownian motion
  - $t \in [0, +\infty]$

Forward SDE



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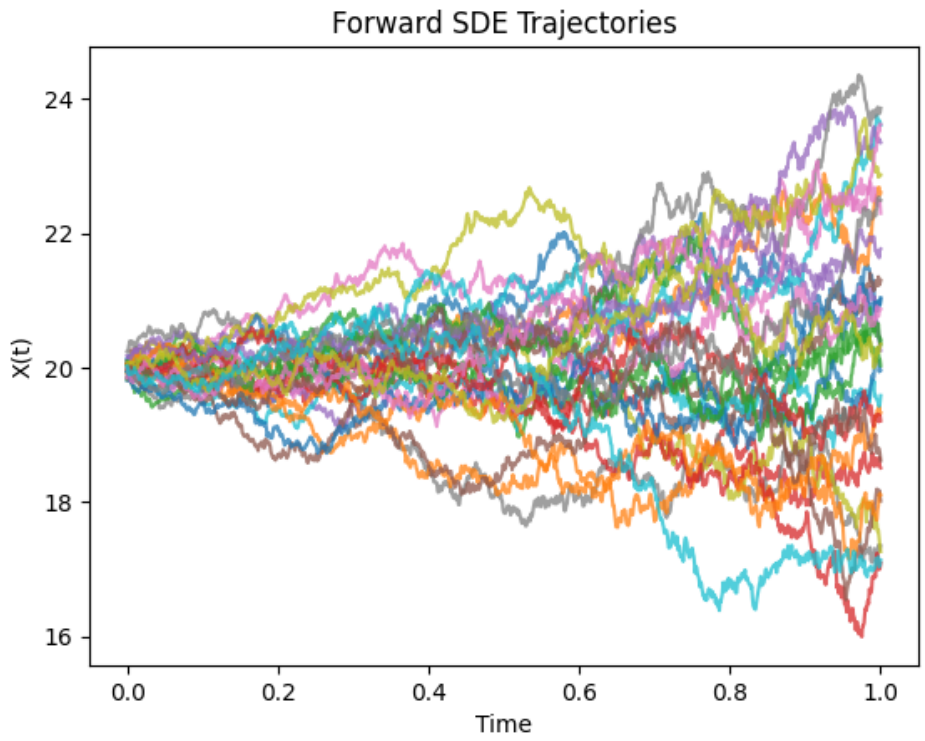

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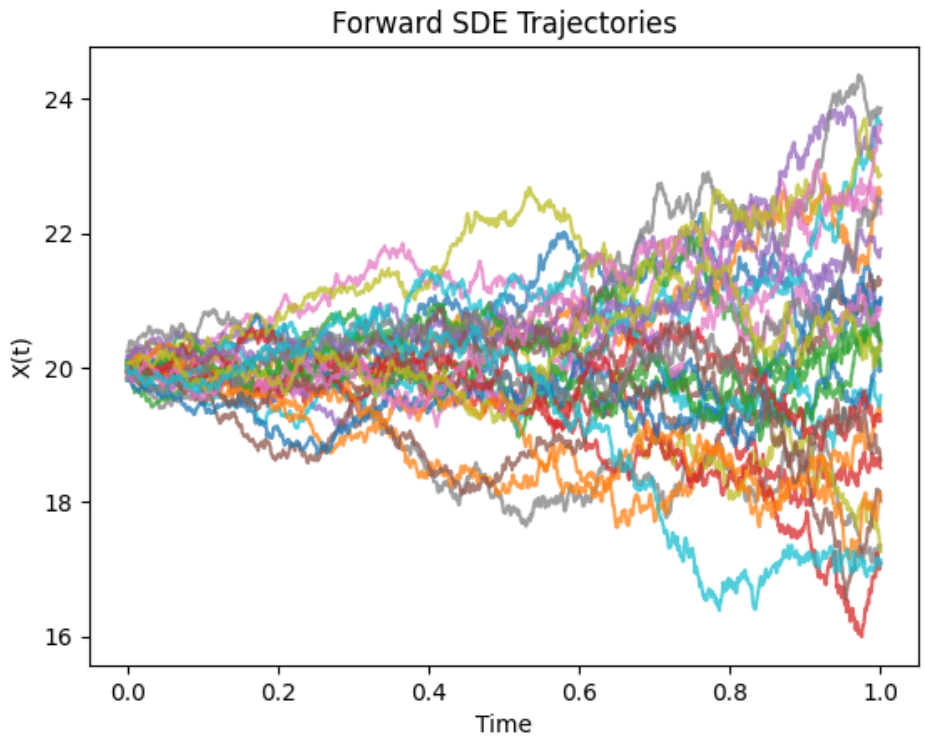

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- Let $p_t(x)$ denote the (marginal) probability density function of $x(t)$.
- $p_0(x)$ is the data distribution and $p_T(x)$ is tractable noise distribution

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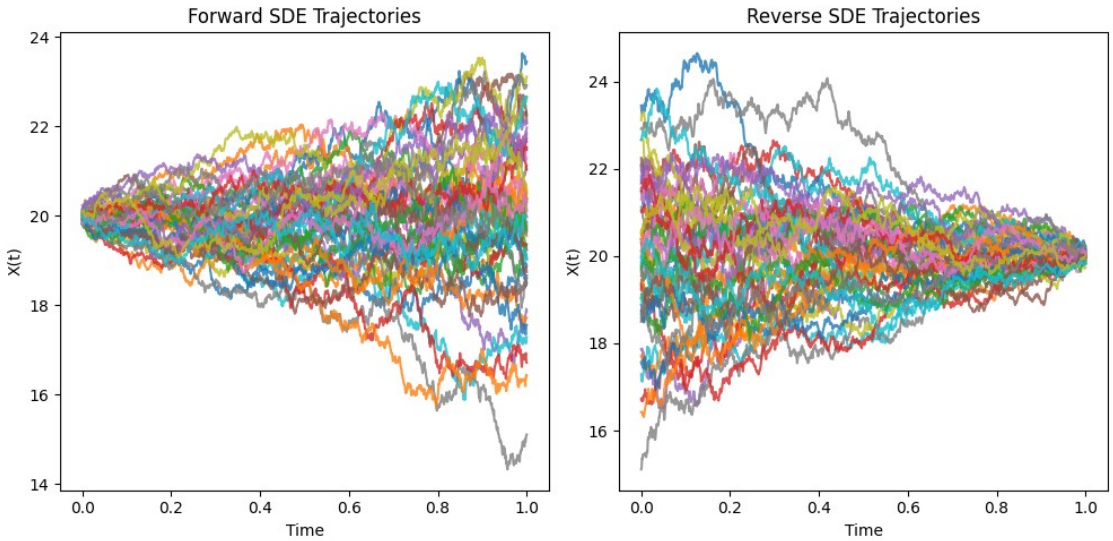

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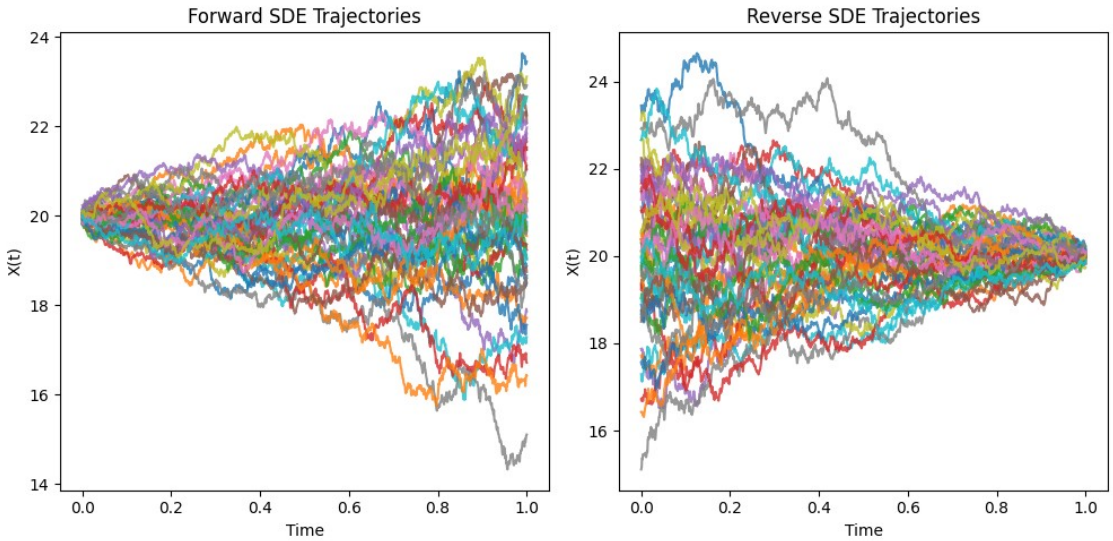

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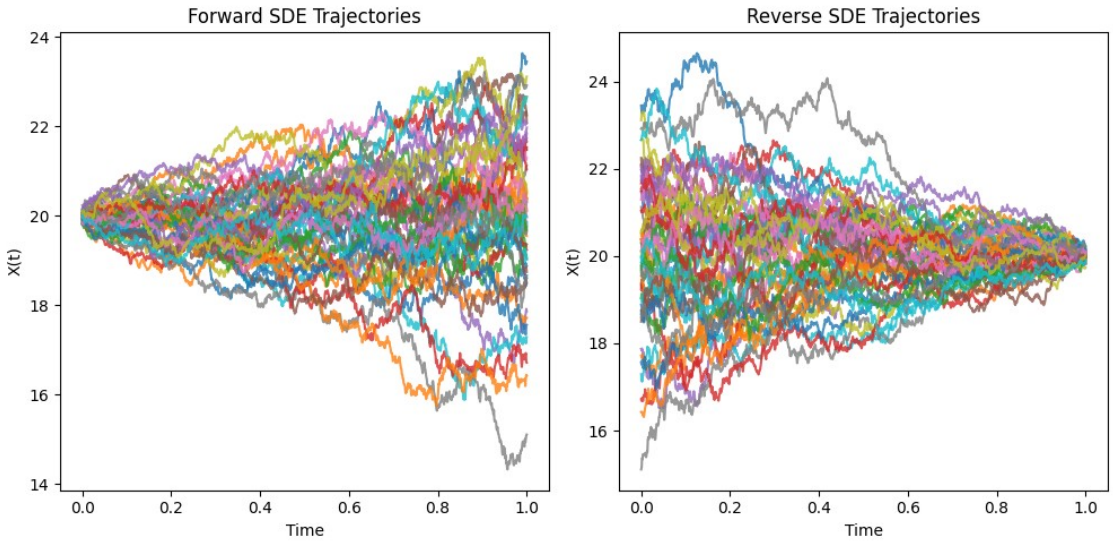

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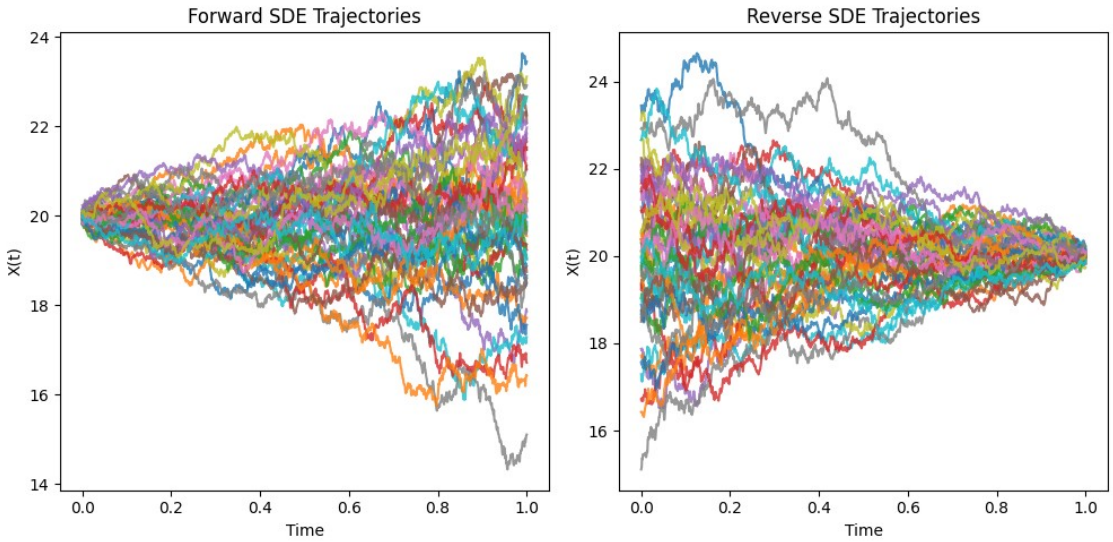

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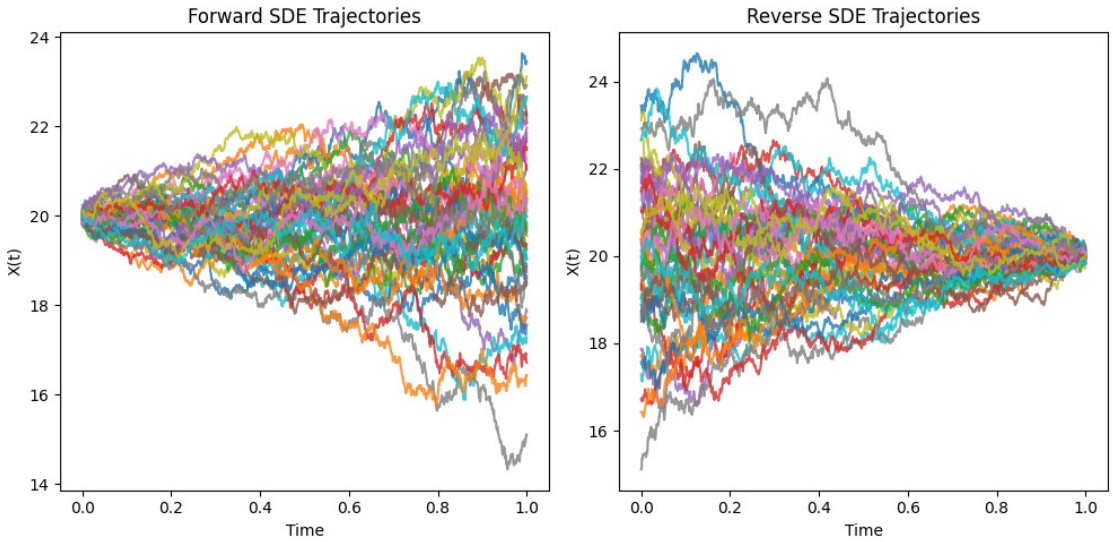

Target Distribution: Gaussian Mixture Model

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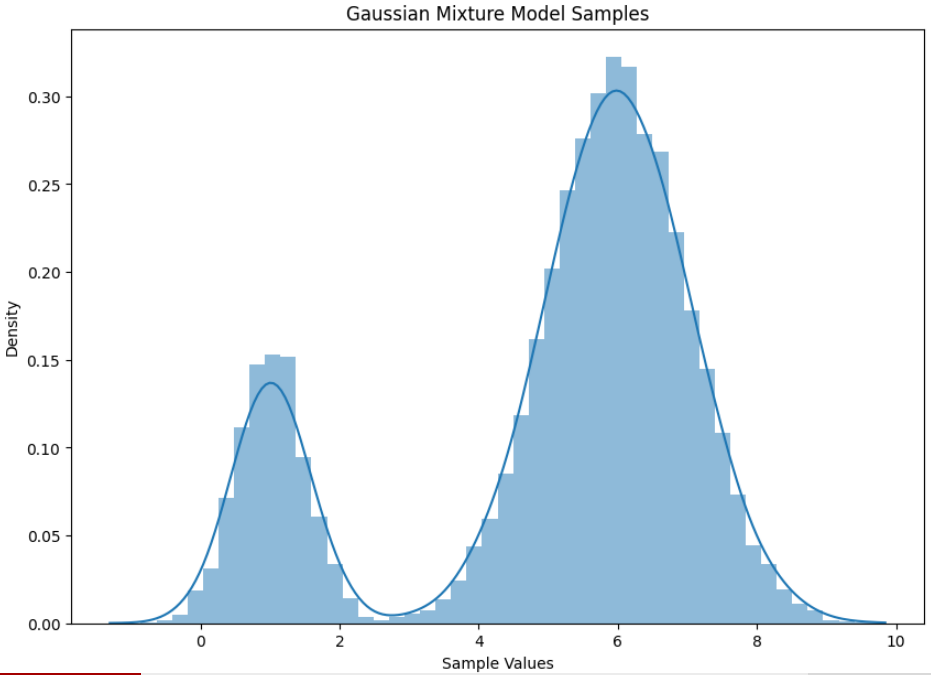

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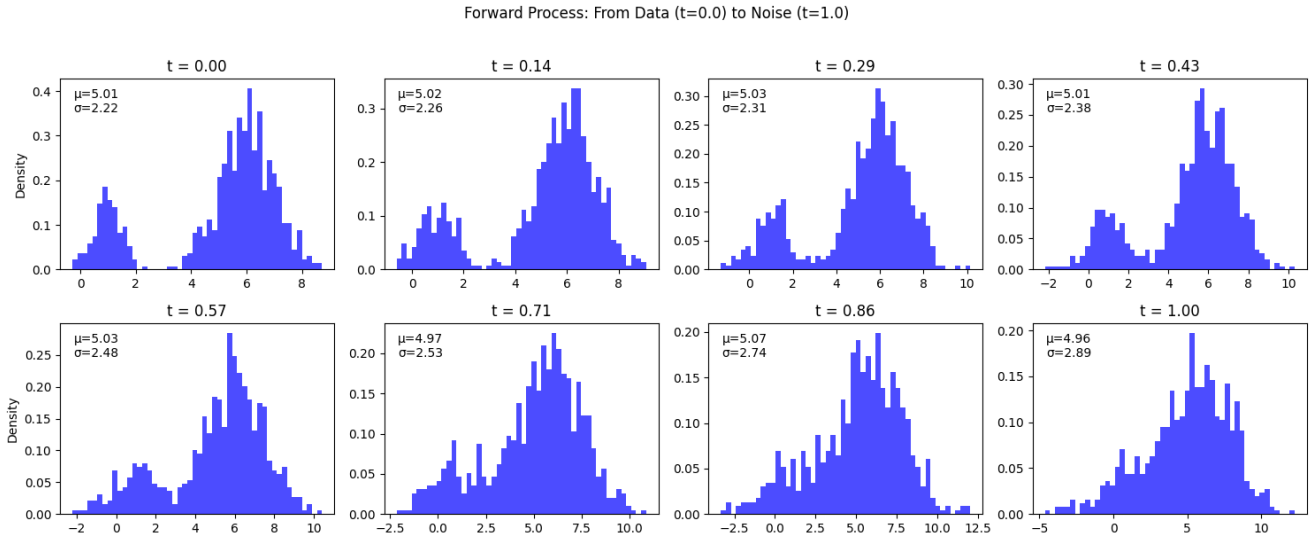

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- Where our neural network was trained to minimize:

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Score Function Estimation

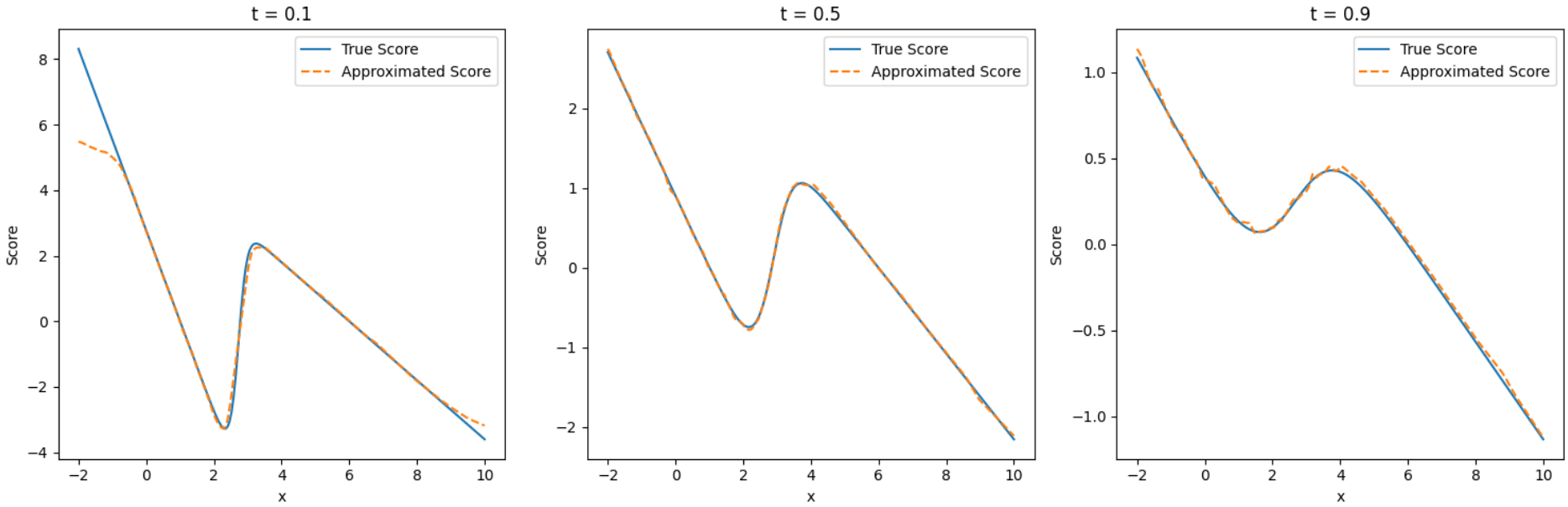

Figure: Comparison of true score vs. learned score at different time steps

Data Recovery Results

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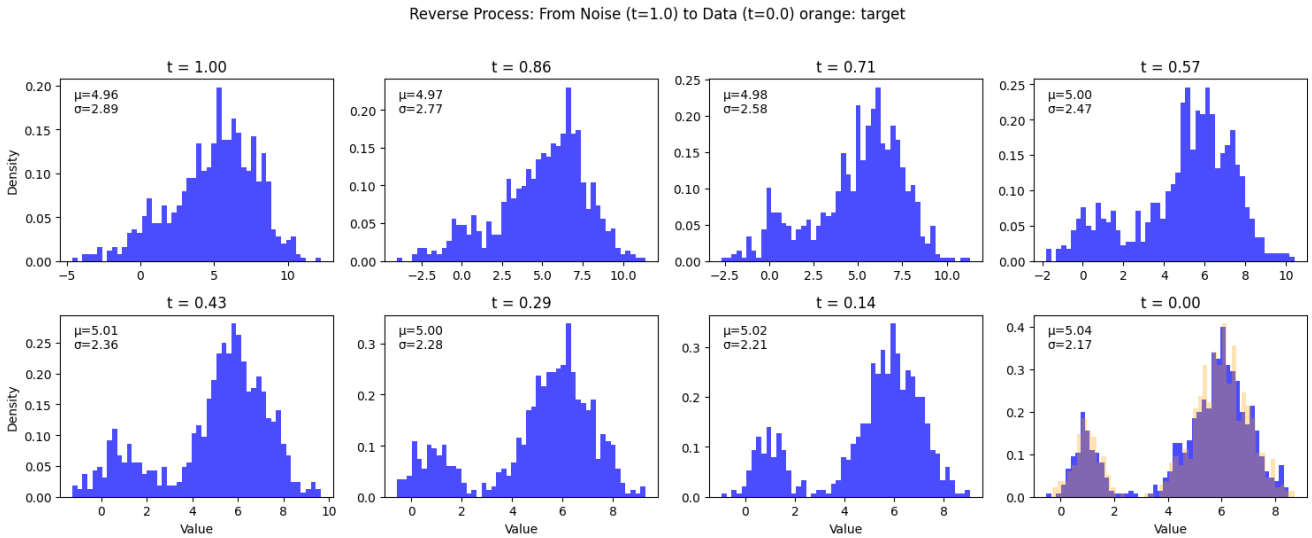

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where C is a constant independent of θ and $\nabla_{\mathbf{x}}\mathbf{s}_m$ is the Hessian

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where C is a constant independent of θ and $\nabla_{\mathbf{x}}\mathbf{s}_m$ is the Hessian

- This gives us an unbiased estimator:

$$\hat{J}(\theta;\mathbf{x}_1^N) = \frac{1}{N}\sum_{i=1}^{N}\left[\text{tr}(\nabla_{\mathbf{x}}\mathbf{s}_m(\mathbf{x}_i;\theta)) + \frac{1}{2}\|\mathbf{s}_m(\mathbf{x}_i;\theta)\|_2^2\right]$$

Practical Implementation of Score Matching

- **Challenge**: No access to $\mathbf{s}_d(\mathbf{x})$ directly, only samples from p_d
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- Computational challenge: Computing the trace of Hessian requires $O(D)$ backward passes

Pizza time!

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- This only requires Hessian-vector products - computationally efficient!

Let's try this again

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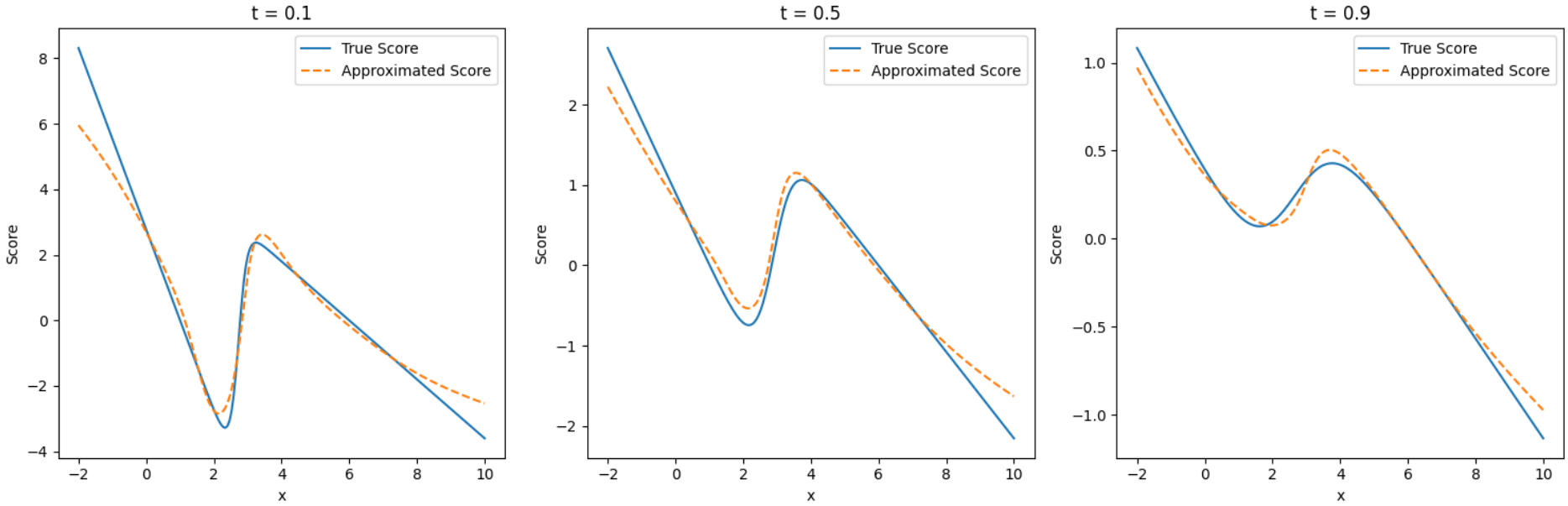
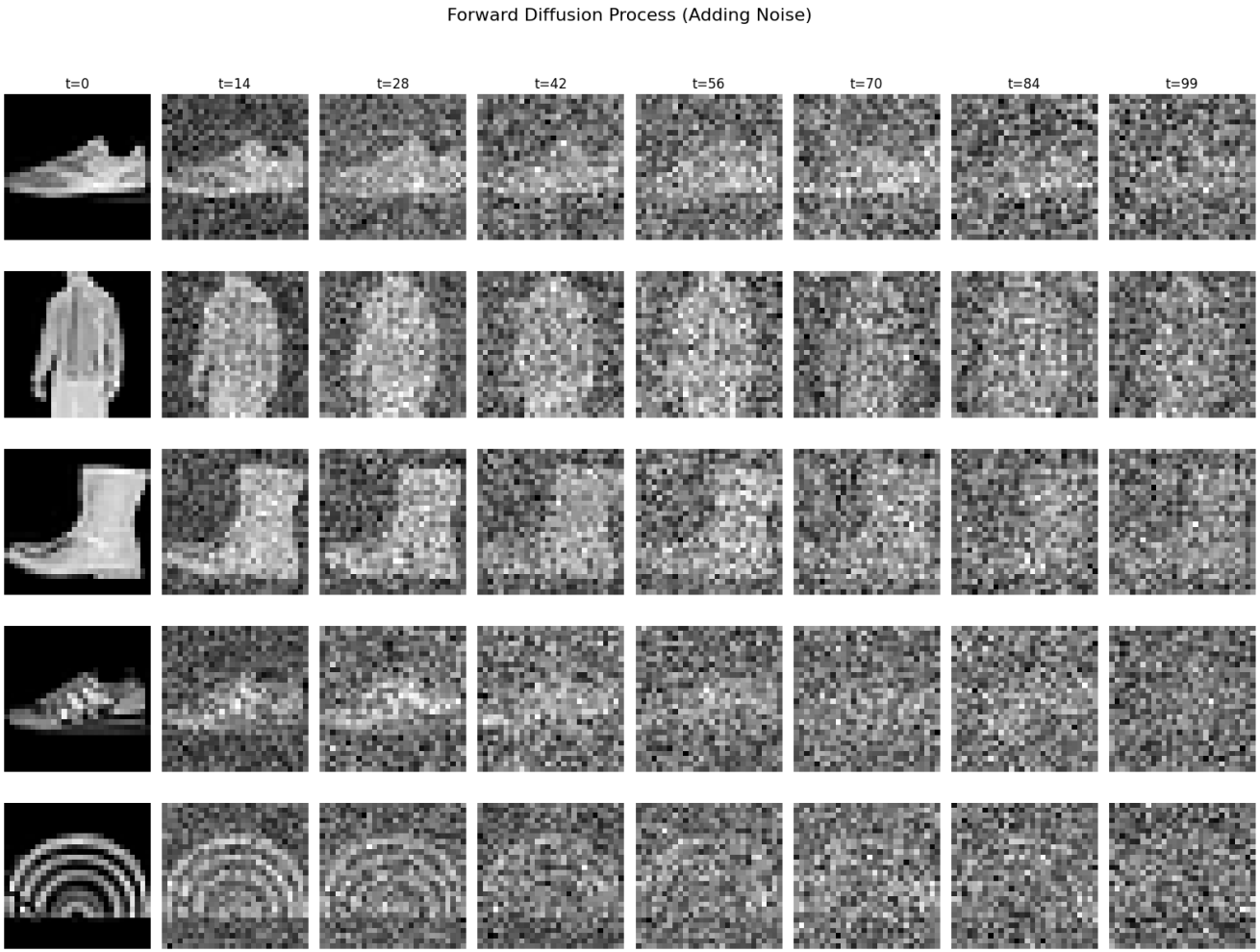


Figure: Comparison of true scores and approximate scores via sliced score matching

Higher dimensions?



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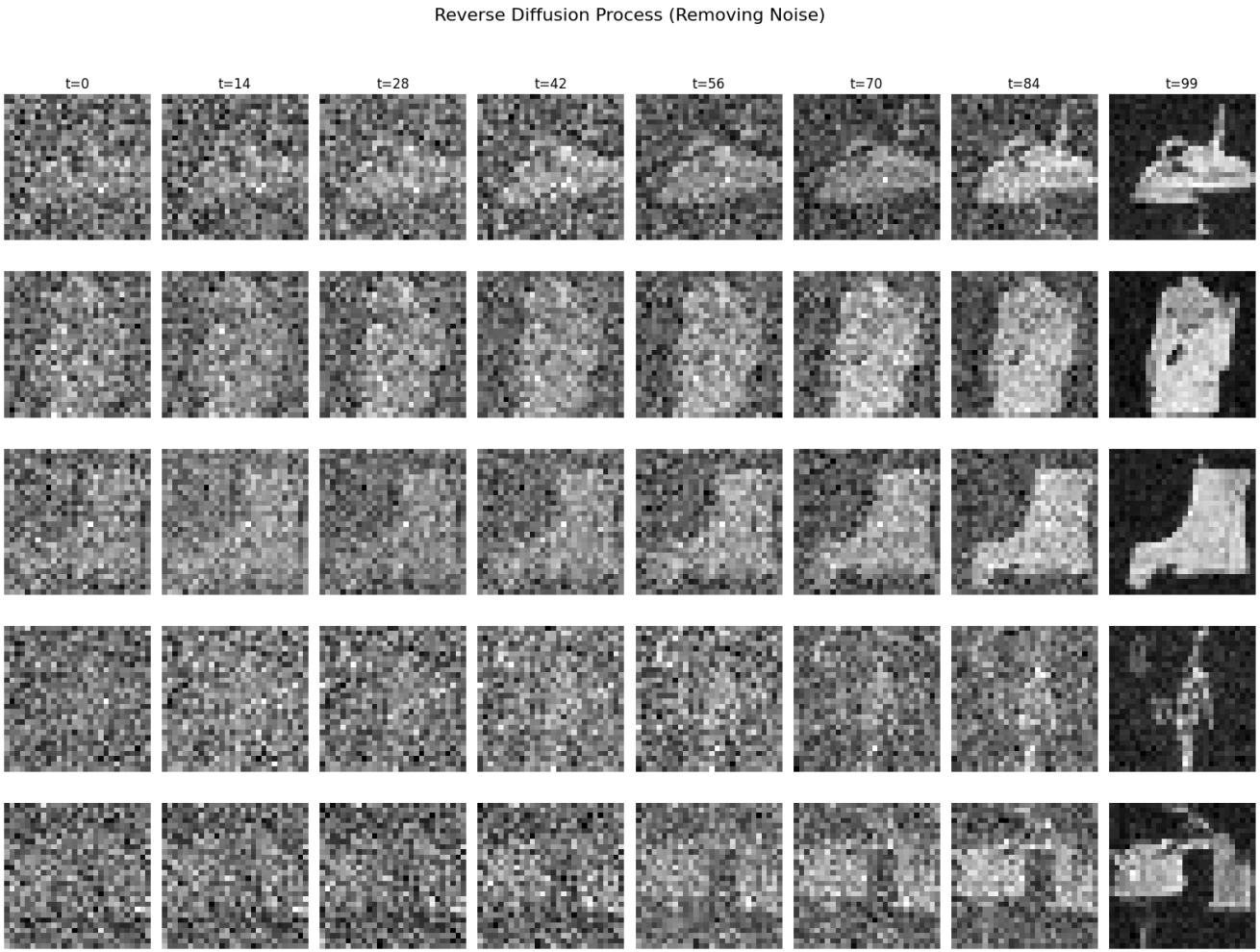

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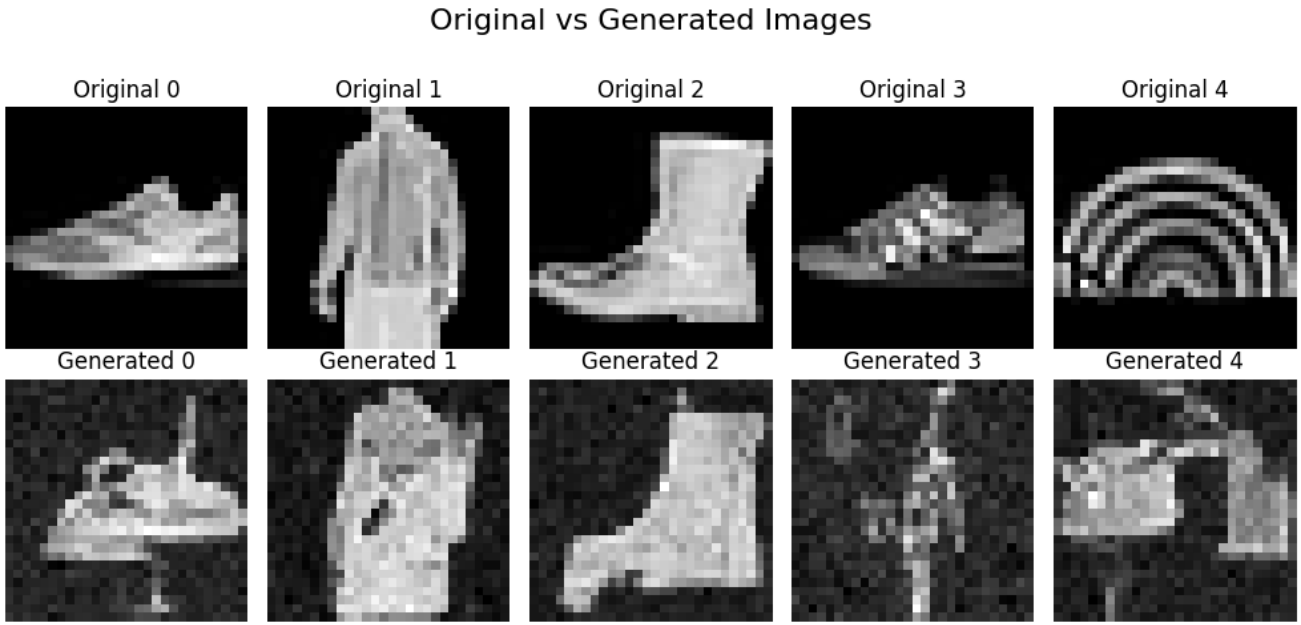

Supercharged Score-Matched Image

Figure: Generated images using U-Net architecture

Are we there yet?

THE END.

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Questions?

More generated images

(a) ODE Sampler

(b) Euler-Maruyama Sampler

(a) Predictive-Corrector Sampler

Practical Implementation of Sliced Score Matching

dataset $\mathbf{x}_1^N$, we draw M random projection vectors $\mathbf{v}_{ij}$ for each $\mathbf{x}_i$

Algorithm 0: Sliced Score Matching

- **Input:** $\tilde{p}_m(\cdot;\theta)$, $\mathbf{x}$, $\mathbf{v}$

  $\mathbf{s}_m(\mathbf{x};\theta) \leftarrow \text{grad}(\log \tilde{p}_m(\mathbf{x};\theta), \mathbf{x})$

  $\mathbf{v}^\top \nabla_{\mathbf{x}} \mathbf{s}_m(\mathbf{x};\theta)\mathbf{v} \leftarrow \text{grad}(\mathbf{v}^\top \mathbf{s}_m(\mathbf{x};\theta), \mathbf{x})$

  $J \leftarrow \frac{1}{2}(\mathbf{v}^\top \mathbf{s}_m(\mathbf{x};\theta))^2$

  $J \leftarrow J + \mathbf{v}^\top \nabla_{\mathbf{x}} \mathbf{s}_m(\mathbf{x};\theta)\mathbf{v}$

  return J

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- Hard to select optimal noise level without prior knowledge

this again?

- Gaussian mixture distribution scores using DSM

¡2-¿

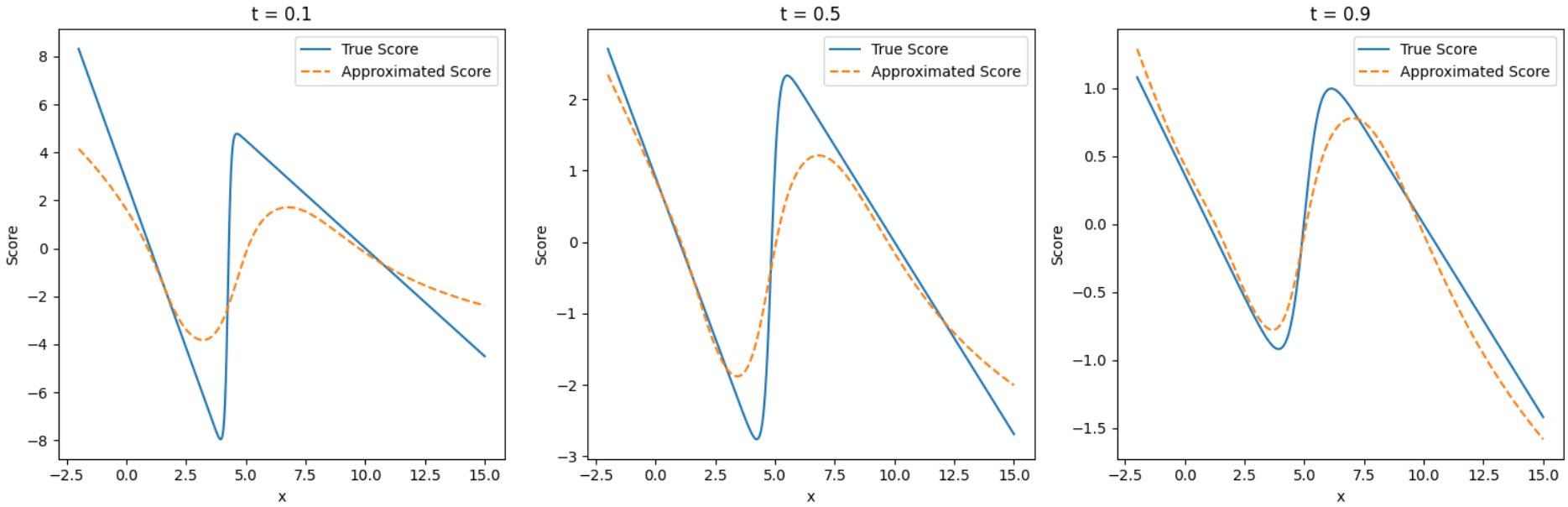

Figure: Comparison of True scores and approximate scores via denoised score matching